

Sensors & Instrumentation

Final Project Proposal

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1 Overview

Students will work in teams of 3–4 members to design, implement, test, and demonstrate an Arduino-based smart sensing system addressing a real-world engineering problem. Each project must integrate at least five sensor modalities — optical, force, flow, temperature, and humidity — and deliver meaningful data acquisition, processing, and visualization.

The project spans three weeks and culminates in a live demonstration and a technical report. It is designed to consolidate course concepts and develop professional skills required for sensor-driven product development.

2 Learning Objectives

Upon successful completion of this project, students will be able to:

- Select and justify appropriate sensors for a given real-world application.
- Interface multiple sensor types with an Arduino microcontroller using digital, analog, I²C, and SPI protocols.
- Implement signal conditioning, calibration routines, and noise-reduction techniques.
- Acquire, log, and visualize sensor data in real time.
- Integrate five or more heterogeneous sensors into a coherent embedded system.
- Evaluate system performance against defined specifications.
- Communicate technical outcomes through a written report and live demonstration.

3 Required Sensor Modalities

Every project must incorporate all five of the following sensor categories. Groups may add additional sensors to enhance their scenario.

Modality	Example Sensors	Measured Quantity	Interface
Optical	LDR, IR photodiode, TCRT5000	Light intensity, object detection	Analog / digital GPIO
Force	FSR 402, HX711 + load cell	Weight, applied force, pressure	Analog / SPI (HX711)
Flow	YF-S201, FS300A	Liquid flow rate (L/min)	Digital pulse / interrupt
Temperature	DS18B20, DHT22, LM35	Ambient or fluid temperature (°C)	1-Wire / Analog / I ² C
Humidity	DHT22, SHT31, BME280	Relative humidity (%RH)	Digital 1-Wire / I ² C

4 Suggested Project Ideas

The following scenarios naturally motivate the use of all five sensor modalities. Groups are encouraged to propose their own scenario subject to instructor approval.

- Smart greenhouse.
- Smart weather station.
- Water quality monitoring system.
- Smart energy monitoring system.
- Industrial safety monitoring platform.
- Smart building environmental monitor.
- Cold-chain logistics monitor.
- Industrial fluid pipeline monitor.

Groups wishing to propose a custom scenario must submit a one-page description to the instructor by the end of Day 2 of Week 1. Approval requires demonstrating that all five sensor modalities serve a meaningful and distinct role within the scenario.

5 System-Level Functional Requirements

In addition to acquiring sensor data, each project must implement a complete sensing system capable of **monitoring, analyzing, and responding to environmental conditions.**

Every project must:

- Acquire data simultaneously from all required sensor modalities.
- Detect at least one abnormal or threshold-based condition.

- Trigger at least one automatic response based on sensor readings.
- Log all measurements with timestamps.
- Provide real-time visualization of system status.
- Operate continuously for a minimum of 60 seconds during the live demonstration.

Examples of automatic responses include: activating an LED or buzzer alarm, controlling a relay or actuator, sending a warning message to a dashboard, or switching operating modes automatically.

6 Calibration Requirements

Sensor calibration is a mandatory component of this project. Each sensor must be calibrated using at least three known reference points.

The calibration section of the report must include:

- Raw sensor measurements.
- Reference values.
- Calibration curve or equation.
- Calibration procedure.
- Measurement error before and after calibration.

Projects that do not include calibration evidence will lose marks in the Data Analysis and Technical Quality evaluation categories.

7 Measurement Uncertainty and Error Analysis

Instrumentation systems are never perfectly accurate. The final report must discuss:

- Sensor accuracy and resolution.
- Repeatability.
- Sources of measurement noise and environmental influences.
- Possible improvements to measurement quality.

Whenever possible, numerical error values should be calculated and presented.

8 Engineering Constraints

The project shall demonstrate sound engineering design principles. Each group must discuss the following constraints and explain how they influenced design decisions:

- Cost considerations.

- Reliability and robustness.
- Electrical safety (maximum operating voltage: 12 V DC; no direct connection to mains electricity; proper protection for pumps, motors, and relays).
- Maintainability and power consumption.
- Scalability and future improvements.

9 Technical Requirements

Hardware

- Arduino Uno, Mega 2560, or compatible board (ESP32 permitted with instructor approval).
- One sensor per required modality; components listed in the approved Bill of Materials (BOM).
- Appropriate signal conditioning: voltage dividers, pull-up resistors, op-amp buffers where required.
- Wiring documented on a schematic produced with Fritzing, KiCad, or equivalent.

Firmware

- Code written in Arduino C/C++ and managed in a GitHub repository.
- Modular structure: one .ino or .cpp/.h file per sensor driver.
- Sampling rate clearly defined and justified for each sensor.
- Calibration constants stored in EEPROM or a dedicated constants file; not hard-coded inline.
- Serial output formatted as CSV for easy import into analysis tools.

Data & Visualization

- Minimum 60 seconds of continuous multi-sensor data logged during the demonstration.
- At least one real-time display: LCD/OLED on the device, or live serial plotter / Python script.
- Data exported and plotted in Excel, MATLAB, Python (matplotlib), or equivalent.

10 GitHub Repository Structure

README.md
/docs
/code

/data
/video
/report

11 Technical Report Structure

1. Introduction
2. Problem Statement
3. System Architecture
4. Sensor Selection and Justification
5. Hardware Design and Schematics
6. Software Design and Firmware Architecture
7. Calibration Procedures and Results
8. Experimental Results and Data Analysis
9. Measurement Uncertainty and Error Analysis
10. Engineering Constraints Discussion
11. Conclusion and Future Work
12. References

12 Project Timeline

Week	Activities
1	Proposal, scenario justification, system architecture, sensor selection, individual sensor breadboard tests
2	Hardware and software implementation, calibration routines, integration, Progress Report
3	System-level testing, error analysis, Final Report, demonstration video, presentation

13 Deliverables & Grading

Deliverable	Weight
Proposal (scenario description, architecture, BOM, team roles, schedule)	10%
Progress Report (sensor tests, calibration progress, integration status)	15%
Final Technical Report	25%
Live Demonstration (working system, Q&A)	30%
Code Repository (documentation, code quality, version control)	10%
Individual Contribution (summary, Git activity, reflection)	10%
TOTAL	100%

14 Individual Contribution Assessment

Although the project is completed in groups, each student will be assessed individually. Each student shall submit:

- A one-page contribution summary.
- Evidence of work performed.
- Git commit history.
- Personal reflection describing lessons learned.

Failure to demonstrate meaningful participation may result in individual grade adjustments.

15 Bonus Features

Projects demonstrating advanced functionality may receive up to 10 bonus marks (awarded at the instructor's discretion). Examples include:

- IoT connectivity and cloud data logging.
- Mobile application integration or wireless monitoring.
- Predictive analytics or fault detection algorithms.
- Machine-learning-based decision support.

16 Presentation Requirements

- 10-minute technical presentation.
- 5-minute live demonstration.
- 5-minute questions and answers.

17 Academic Integrity

- All code, schematics, and written content must be the original work of the group.
- External libraries or code snippets must be credited with author, source URL, and licence in the README.
- Students may use AI tools as development assistants; however, every submitted component must be fully understood and explainable by all team members.
- No sharing of code, schematics, or data between groups.

18 Submission

All deliverables must be submitted through the course GitHub repository before the announced deadline. Report documents must also be uploaded to the course portal as PDF by 23:59 on the respective due date.

Good luck — build something that matters!